UNIT-1

Introduction to Software and Software Engineering

A Layered Technology/software process layers

Software Process Models, The Linear Sequential Model,

The Prototyping Model, Evolutionary Process Models

Introduction to software process:

Generic process framework activities, Umbrella activities.

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SOFTWARE
UX
XML
CSS
<code/>
AI
Ps
UI
C+



What is Software Engineering?

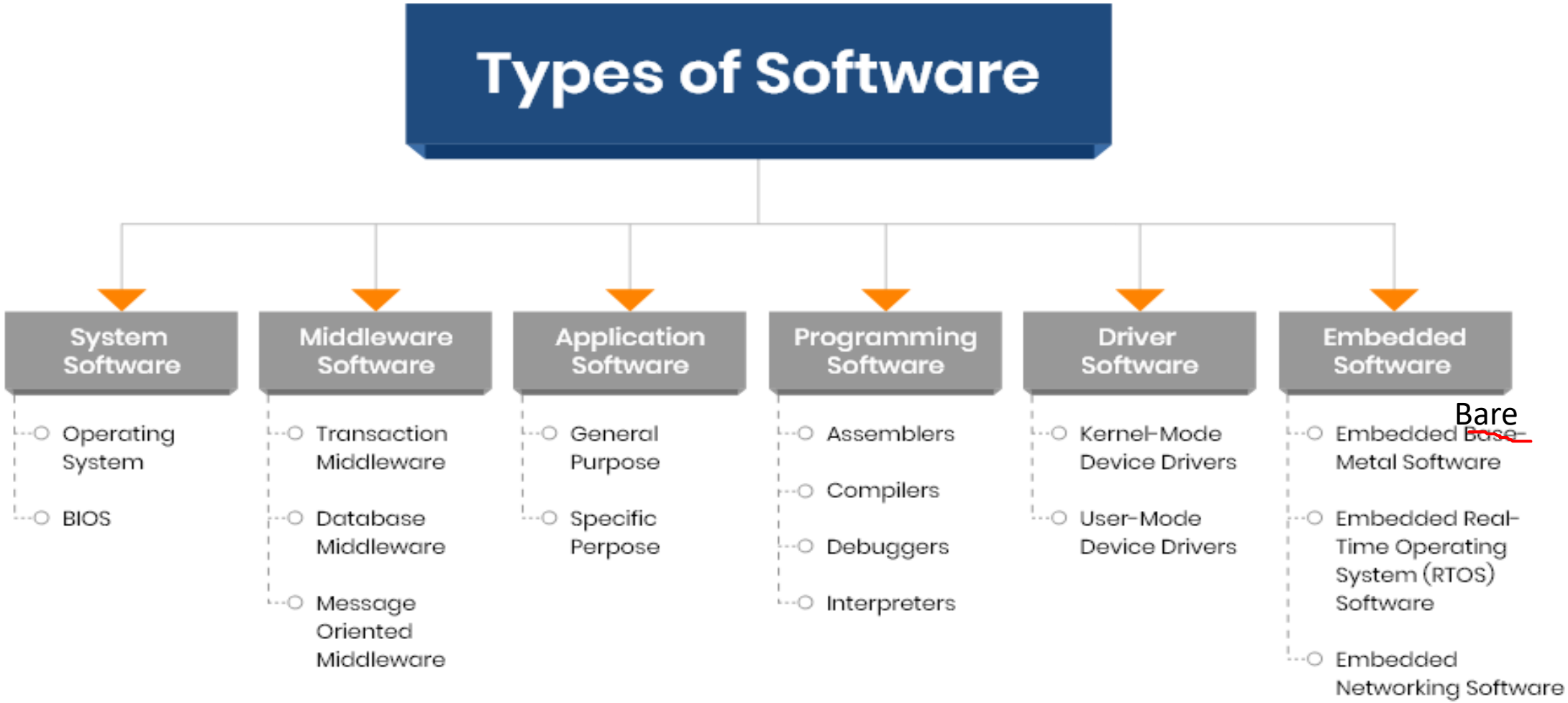
Software engineering is a combination of two terms: software and engineering. So the very first question that strikes our minds is**, What is software?**

- **Software** is a program or set of programs containing instructions that provide the desired functionality.

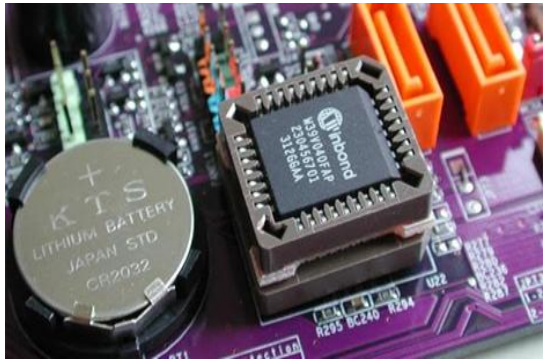
Then the next question is: **What is engineering?**

- **Engineering** is the process of designing and building something that serves a particular purpose and finds a cost-effective solution to problems.

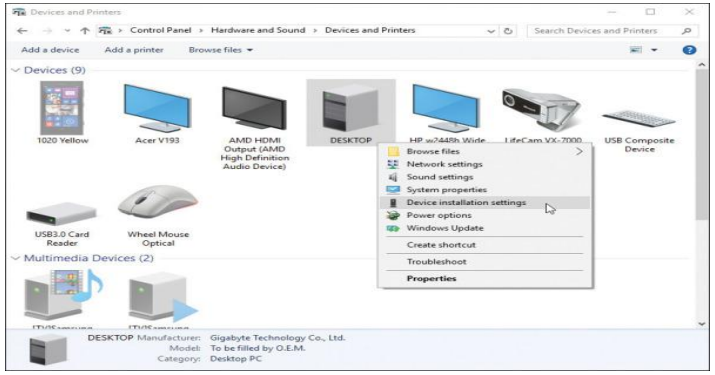
Types of Software



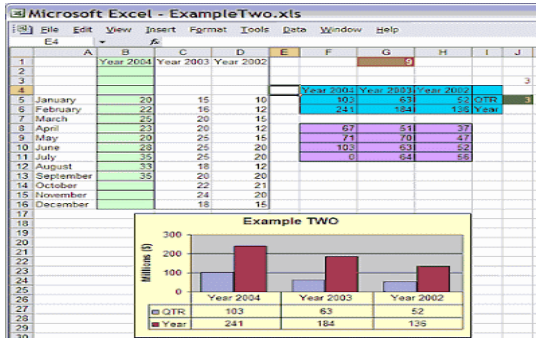
Bare

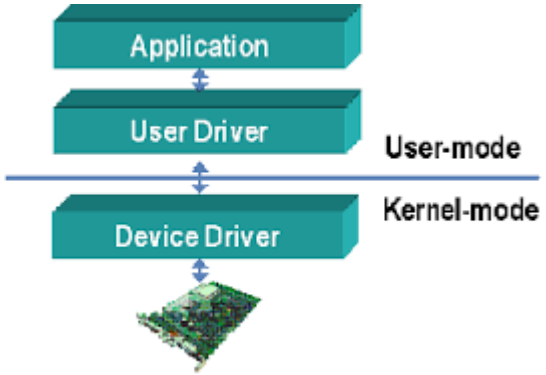












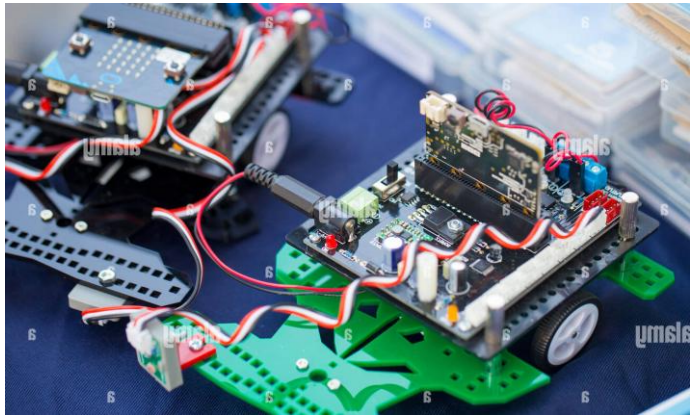
Bare-Metal PROGRAMMING LANGUAGES

IN Tech House



ASSEMBLY
C C++





Software Engineering is the process of designing, developing, testing, and maintaining software.

It is a systematic and disciplined approach to software development that aims to create high-quality, reliable, and maintainable software.

Software engineering includes a variety of techniques, tools, and methodologies, including requirements analysis, design, testing, and maintenance.

A Layered Technology/software process layers



1. A quality focus:

- It defines the continuous process improvement principles of software.
- It provides integrity that means providing security to the software so that data can be accessed by only an authorized person, no outsider can access the data.
- It also focuses on maintainability and usability.

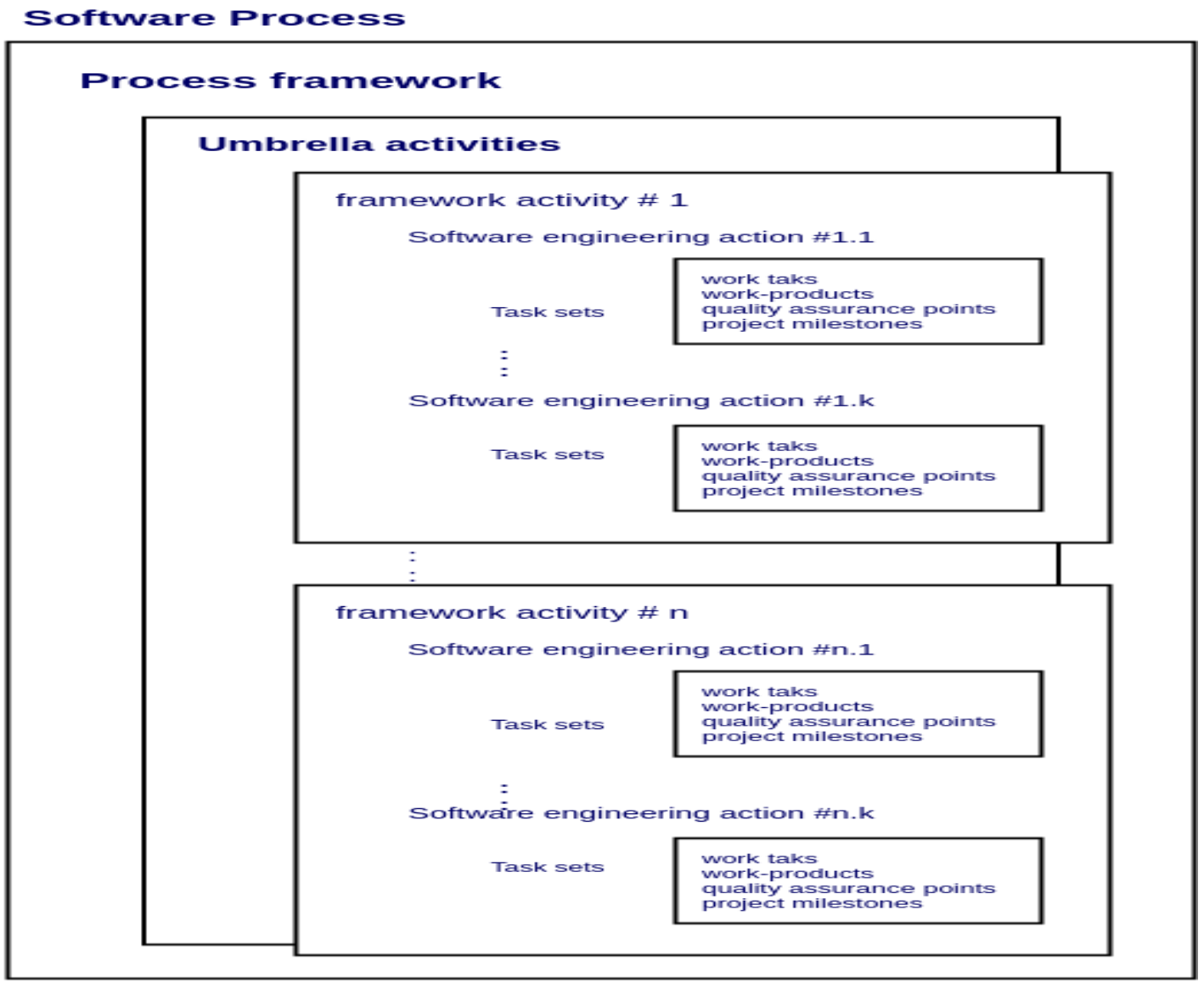
Process activities are listed below:-

- Communication
- Planning
- Modelling
- Construction
- Deployment



2. Process:

- It is the foundation of software engineering.
- It is key that binds all the layers together which enables the development of software before the deadline or on time.
- Process defines a **framework** that must be established for the effective delivery of software engineering technology.
- The software process covers all the activities, actions, and tasks required to be carried out for software development.


Software Process
Process framework
Umbrella activities
framework activity # 1
Software engineering action #1.1
Task sets
work taks
work-products
quality assurance points
project milestones
Software engineering action #1.k
Task sets
work taks
work-products
quality assurance points
project milestones
framework activity # n
Software engineering action #n.1
Task sets
work taks
work-products
quality assurance points
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Software engineering action #n.k
Task sets
work taks
work-products
quality assurance points
project milestones

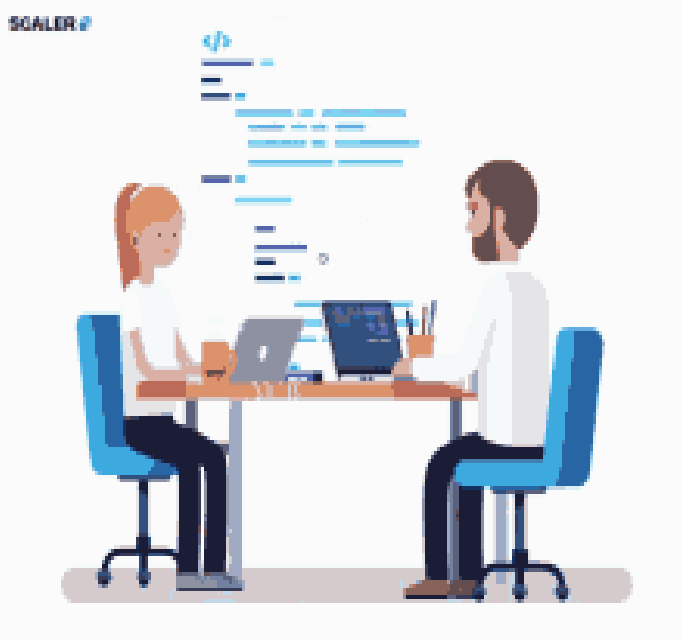
3. Method:

- During the process of software development the answers to all "**how-to-do" questions are given by method.**
- It has the information of all the tasks which includes **communication, requirement analysis, design modeling, program construction, testing, and support.**

4. Tools:

- Software engineering tools provide a self - operating system for processes and methods.
- Tools are integrated which means information created by one tool can be used by another.

Software Process Models



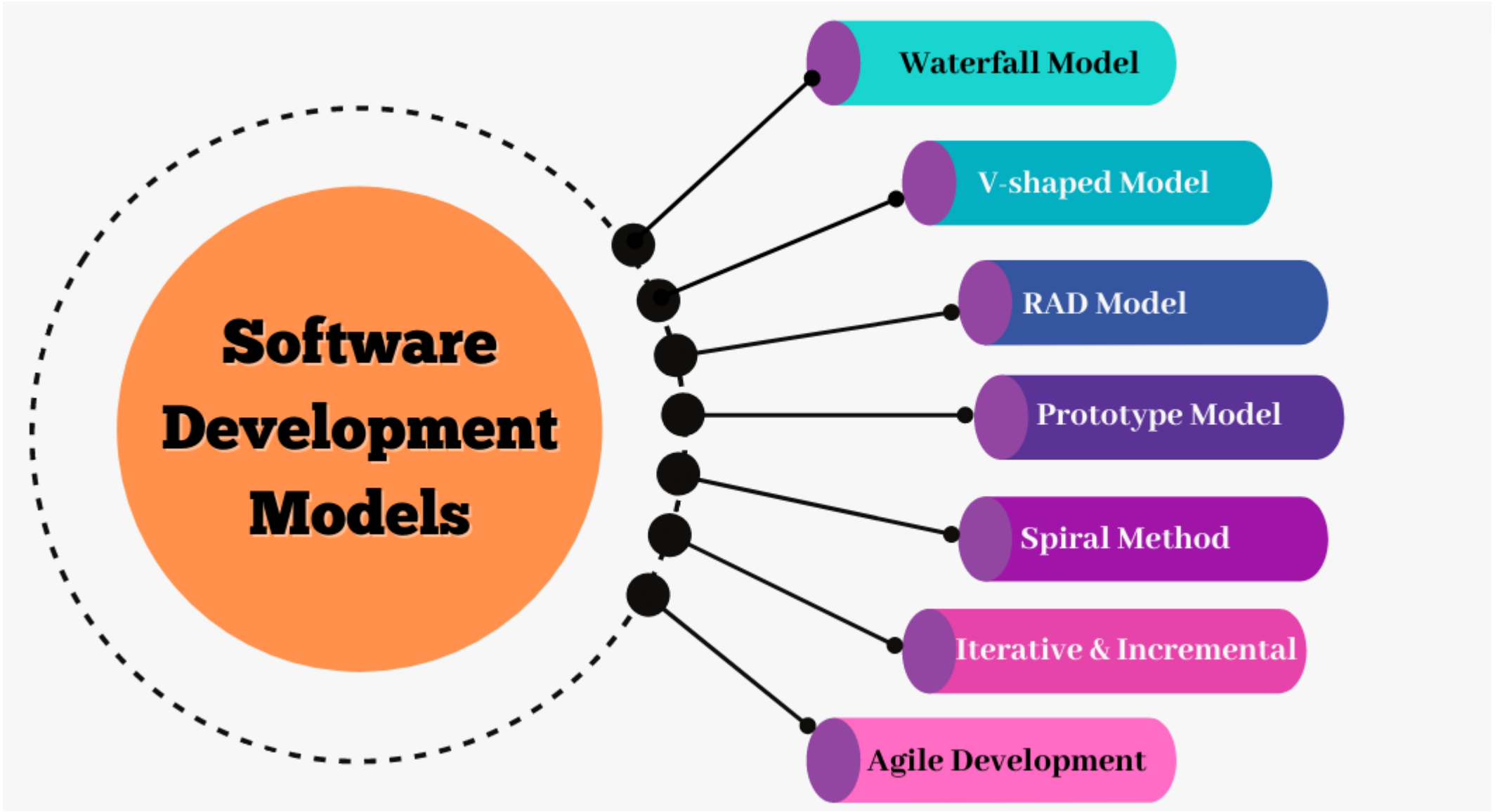


- Software processes are the activities for designing, implementing, and testing a software system.

- The software development process is complicated and involves a lot more than technical knowledge.

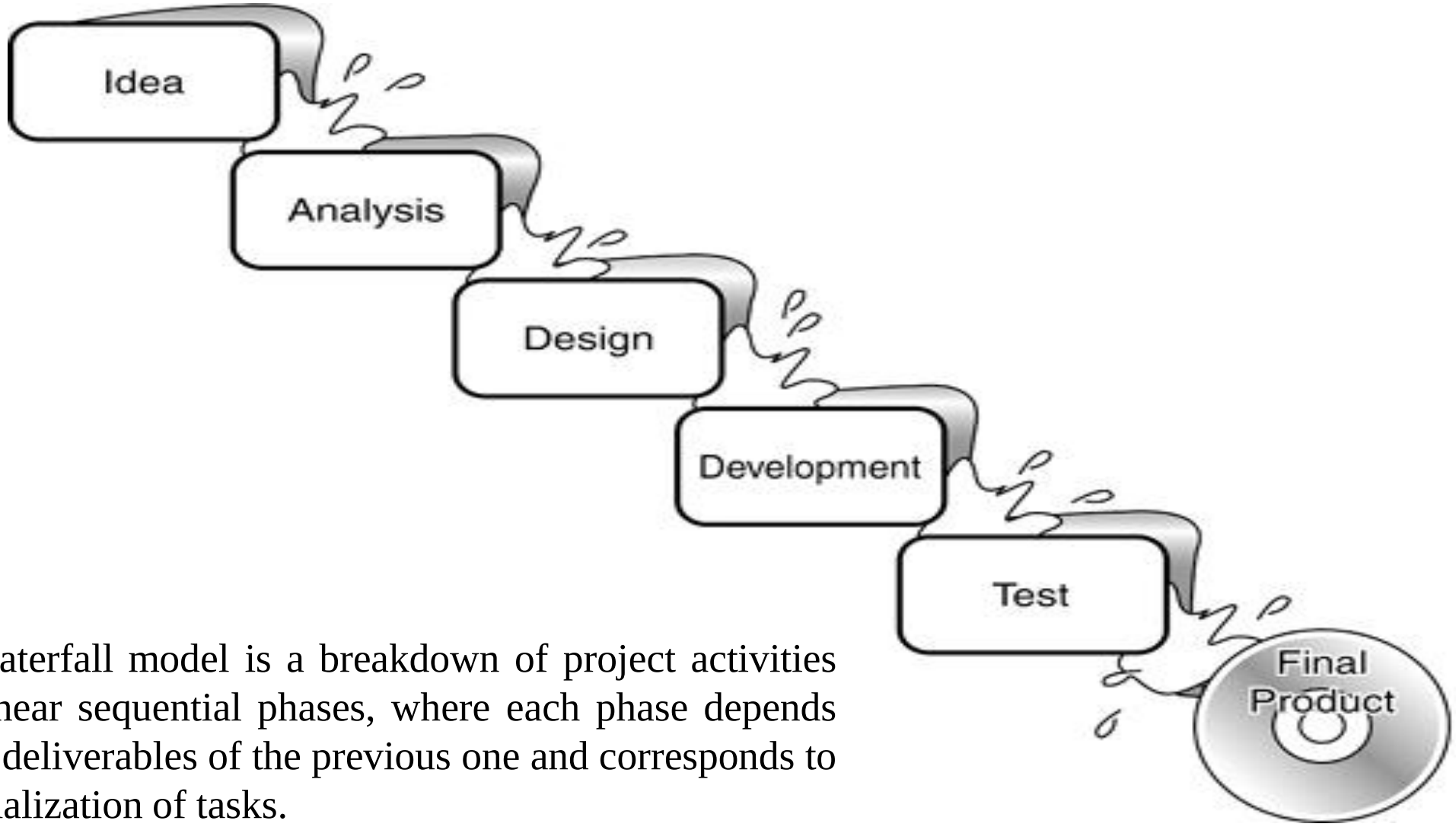
- A software process model is an **abstract representation** of the development process.

- **A model will define the following:**
- The tasks to be performed
- The input and output of each task
- The pre and post conditions for each task
- The flow and sequence of each task


Software Development Models
Waterfall Model
V-shaped Model
RAD Model
Prototype Model
Spiral Method
Iterative & Incremental
Agile Development

➢ Software Process and Software Development Lifecycle Model:

- ➢ Waterfall model
- ➢ V model
- ➢ Incremental model
- ➢ RAD model
- ➢ Agile model
- ➢ Iterative model
- ➢ Spiral model
- ➢ Prototype model



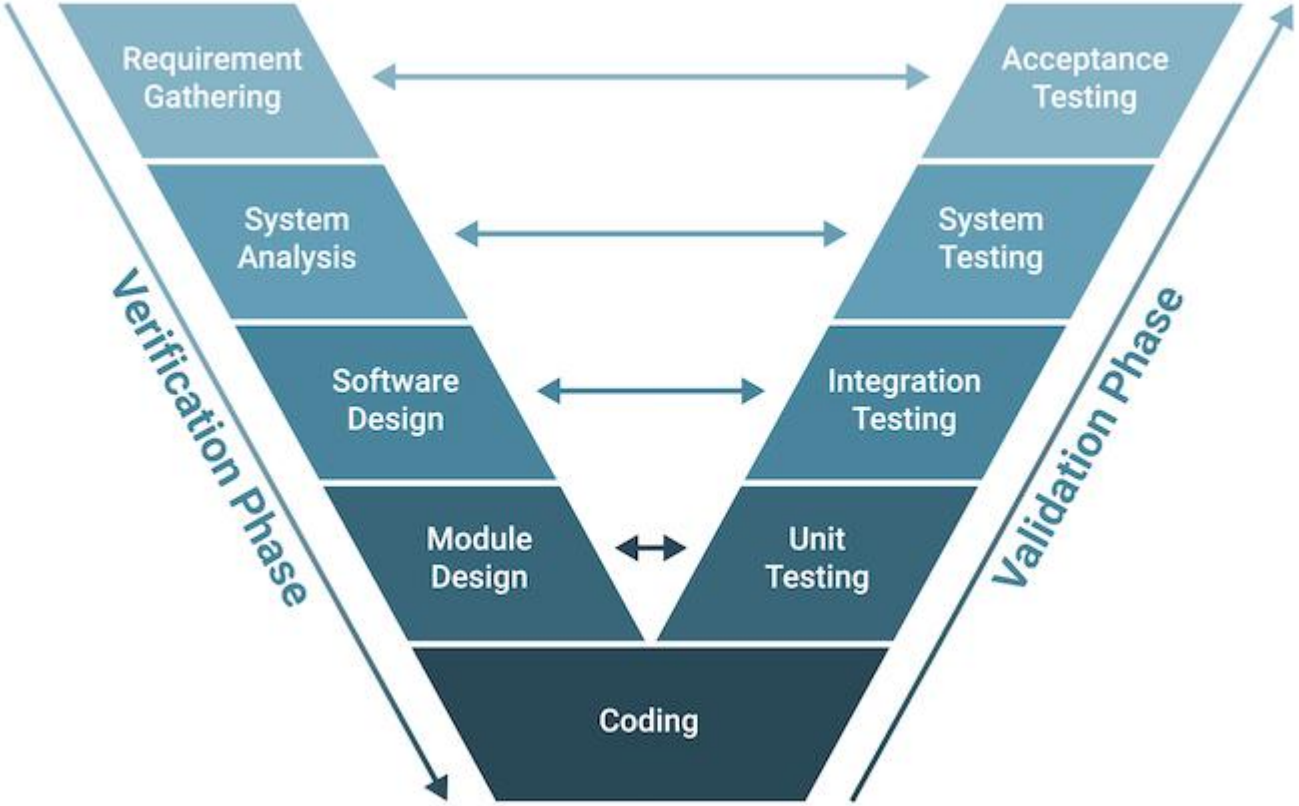
The waterfall model is a breakdown of project activities into linear sequential phases, where each phase depends on the deliverables of the previous one and corresponds to a specialization of tasks.

Advantages and Disadvantages of Waterfall Model

Advantages	Dis-Advantages
•Before the next phase of development, each phase must be completed	•Error can be fixed only during the phase
•Suited for smaller projects where requirements are well defined	•It is not desirable for complex project where requirement changes frequently
•Elaborate documentation is done at every phase of the software's development cycle	•Documentation occupies a lot of time of developers and testers
•Project is completely dependent on project team with minimum client intervention	•Clients valuable feedback cannot be included with ongoing development phase

V-model

- The V-model represents a development process that may be considered an extension of the waterfall model.
- The V-Model demonstrates the relationships between each phase of the development life cycle and its associated phase of testing.
- V-Model also referred to as the Verification and Validation Model.


Requirement Gathering
System Analysis
Software Design
Module Design
Coding
Unit Testing
Integration Testing
System Testing
Acceptance Testing
Verification Phase
Validation Phase

Verification	Validation
Verification is the static process of analyzing documents, visual designs, computer programs, and codes.	Validation is the dynamic process of checking the correct is being built for the user.
It is done by the testers.	It is done by the product team.
The execution of codes is not included in the verification.	The execution of codes is included in the validation.
Verification is done before the validation.	Validation is done after the verification.
Verification checks "Are we building the product right"?	Validation checks "Are we building the right product"?
Verification targets internal aspects such as design.	Validation targets the end product that is ready to be deployed.
It is used to prevent errors.	It is used to detect errors.
Verification testing includes Quality Assurance.	Validation testing includes Quality Control.

The usage of V-Model

- Projects initiated from a request for proposals (RFPs), the customer has a very clear documented requirements
- Military projects
- Mission Critical projects
- Projects with defined and clear requirements

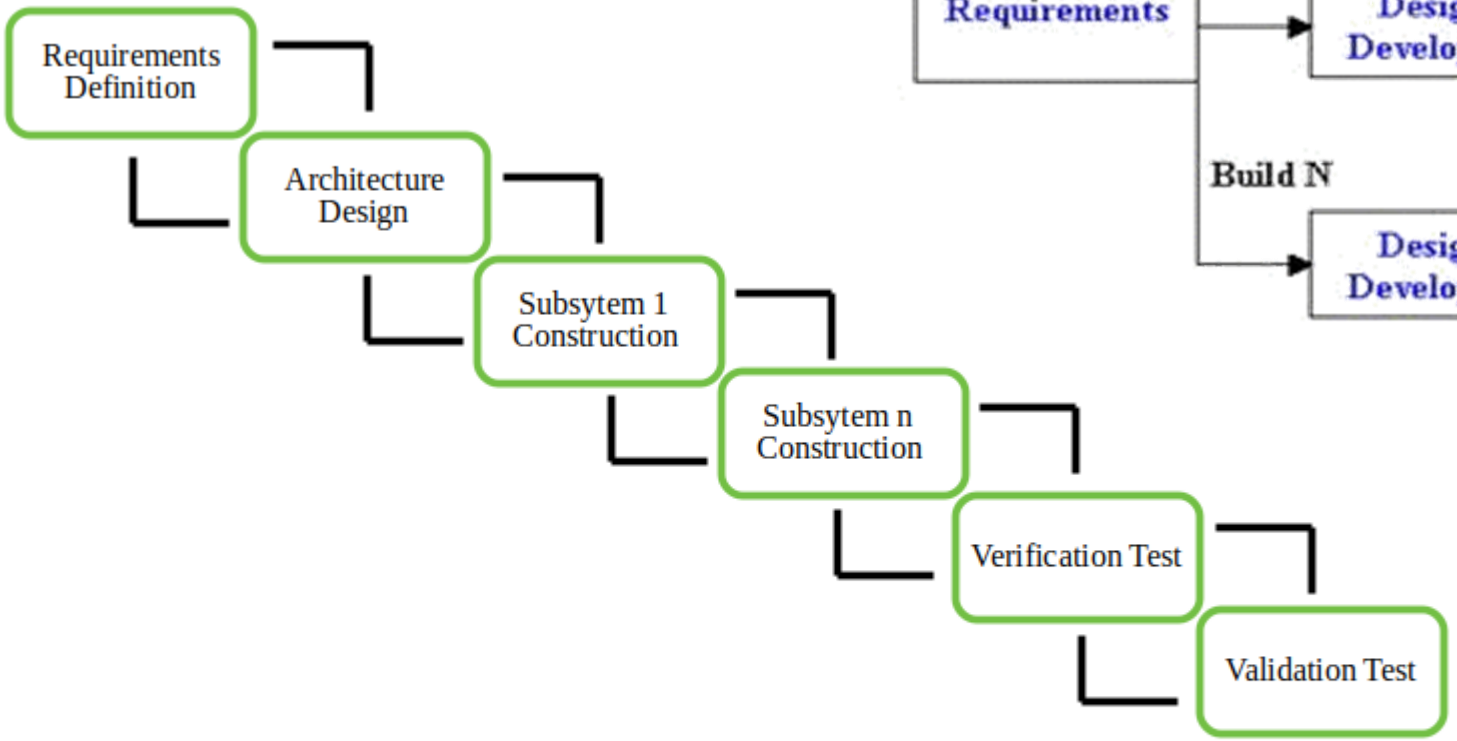
V-Model Model Advantages

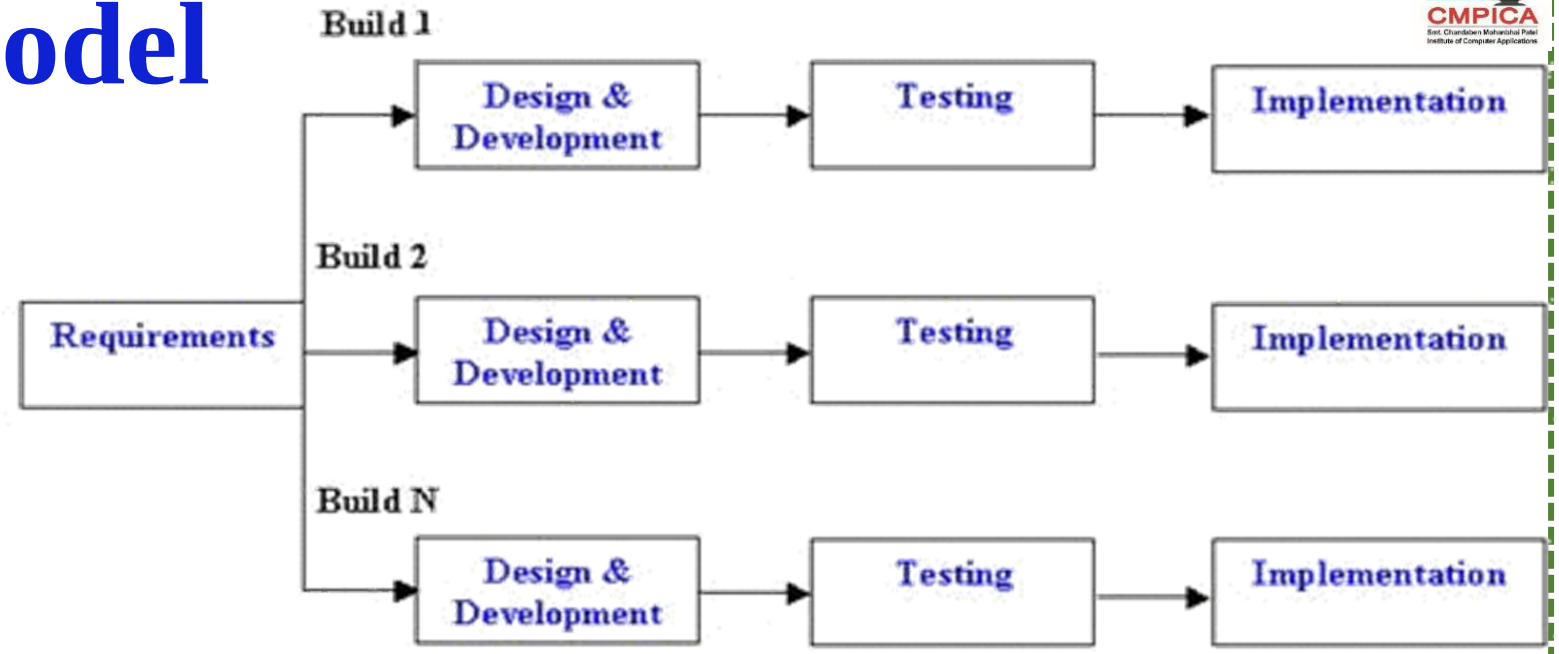
- Simple and easy to use.
- Each phase has specific deliverables.
- Higher chance of success over the waterfall model due to the development of test plans early on during the life cycle.
- Works well for where requirements are easily understood.
- V-Model Improves the quality and reliability of the software.
- It reduces the amount of re-work because of the early detection of defects and issues.

V-Model Model Disadvantages

- Very inflexible, like the waterfall model.
- Adjusting scope is difficult and expensive.
- The software is developed during the implementation phase, so no early prototypes of the software are produced.
- Moreover, It is costly and required more time, in addition to a detailed plan.

Incremental model





Incremental Life Cycle Model

- The incremental build model is a method of software development where the model is designed, implemented and tested incrementally (a little more is added each time) until the product is finished.

- It involves both development and maintenance.

- This model combines the elements of the waterfall model with the iterative philosophy of prototyping.

Advantages of using Incremental process model :

- It is good to use when requirements are known up-front.
- It is good to use when projects having lengthy developments schedules.
- It is good to use when projects uses new Technology.
- It is good to use when **Funding Schedule, Risk, Program Complexity**, or It is more flexible and less costly to change scope and requirements.
- It is easy to manage each iteration.
- Need for early realization of benefits.

Disadvantages of using Incremental process model :

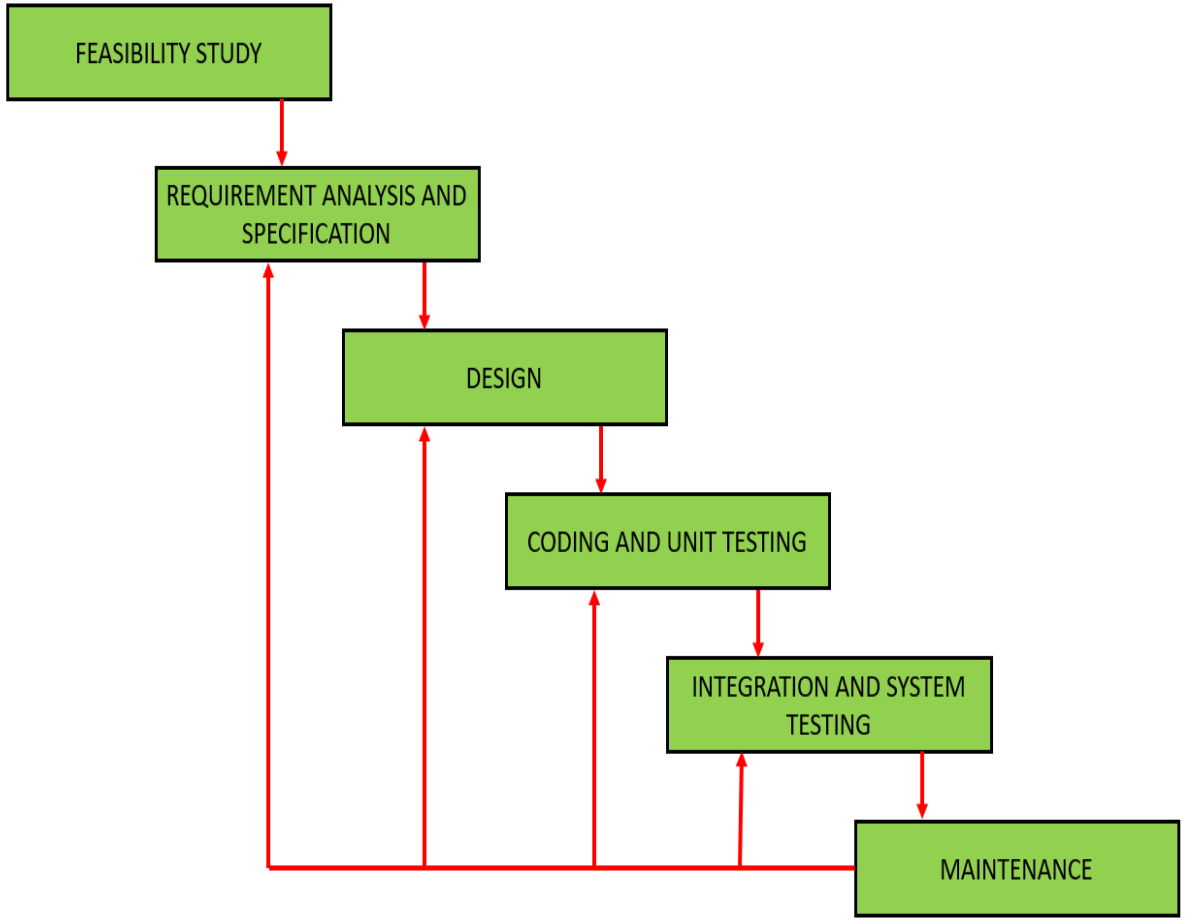
- It requires a good planning designing.
- It is costlier than waterfall model.
- Definition of system should be complete and clear.

Iterative Model

- An iterative life cycle model does not attempt to start with a full specification of requirements by first focusing on an initial, simplified set user features, which then progressively gains more complexity and a broader set of features until the targeted system is complete.

- Major requirements must be defined; however, some functionalities or requested enhancements may evolve with time.

- When adopting the iterative approach, the philosophy of incremental development will also often be used liberally and interchangeably.

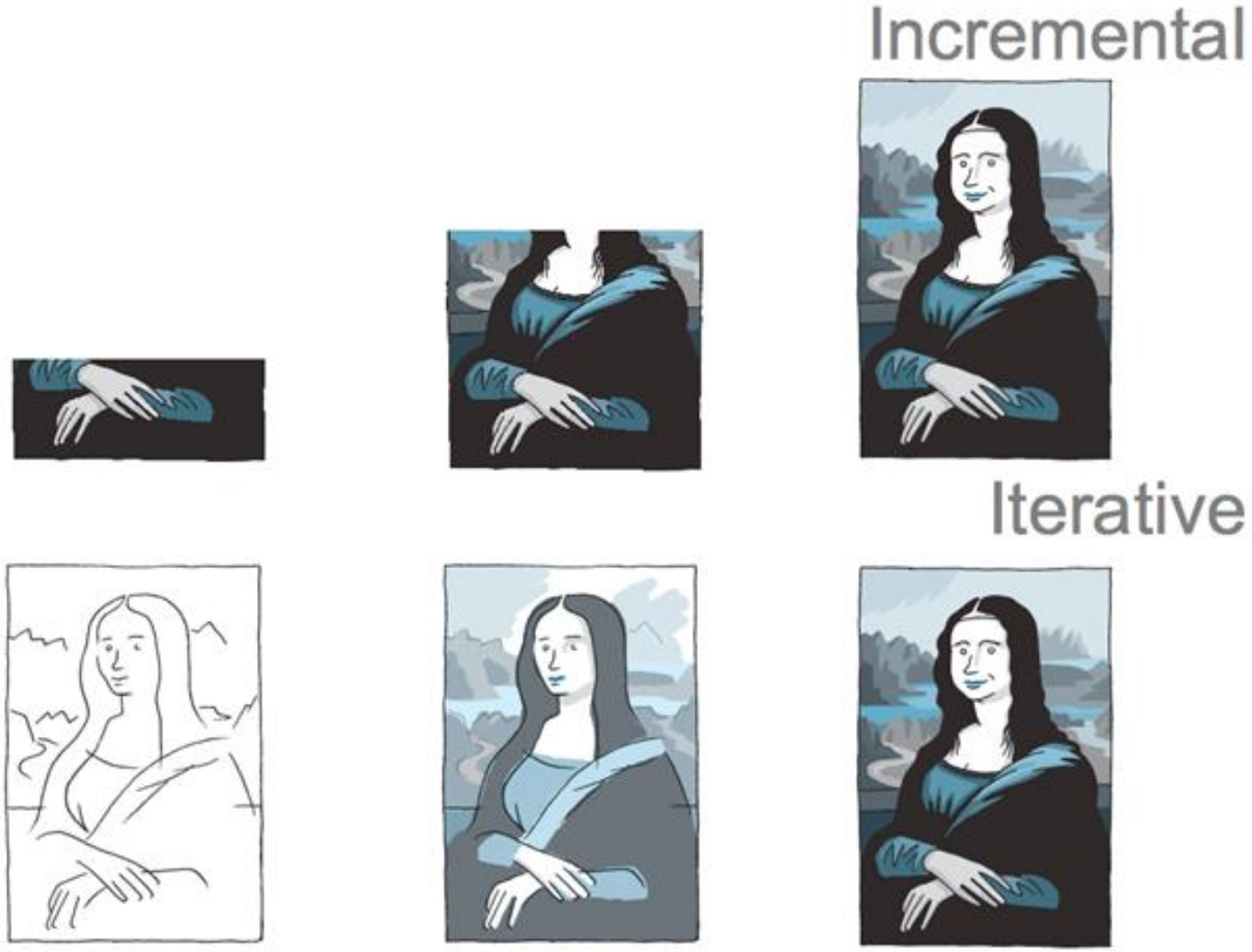

FEASIBILITY STUDY
REQUIREMENT ANALYSIS AND SPECIFICATION
DESIGN
CODING AND UNIT TESTING
INTEGRATION AND SYSTEM TESTING
MAINTENANCE

Advantages of Iterative Model:

- It is efficiently versatile to the forever changeable requirements of the project as well as the customer.
- In iterative model less time is spent on documenting and more time is given for designing.
- Hazards are recognized and fixed through iteration, and every iteration can be simply handled.

Disadvantages of Iterative Model:

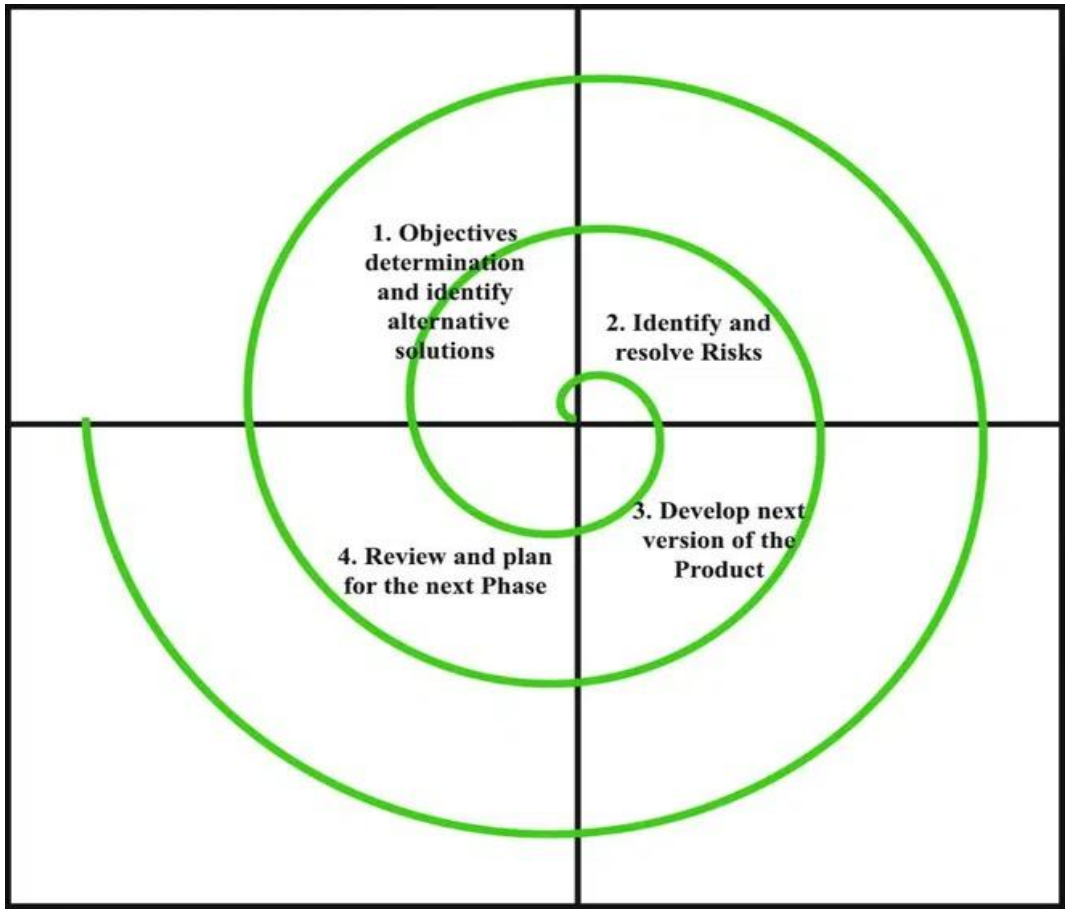
- More Resources may be required.
- Design can be changed again and again because of imperfect requirements.
- Requirement changes can cause over budget.
- Project completion date not confirmed because of changing requirements.
- It is not suitable for smaller projects.


Incremental
Iterative

Spiral model

- A spiral model looks like a spiral with many loops.

- The exact number of loops of the spiral is unknown and can vary from project to project.
- This model supports risk handling, and the project is delivered in loops.
- Each loop of the spiral is called a Phase of the software development process.
- The exact number of phases needed to develop the product can be varied by the project manager depending upon the project risks.


1. Objectives determination and identify alternative solutions
2. Identify and resolve Risks
3. Develop next version of the Product
4. Review and plan for the next Phase

Advantages of Spiral Model:

- Software is produced early in the software life cycle.
- Risk handling is one of important advantages of the Spiral model, it is best development model to follow due to the risk analysis and risk handling at every phase.
- It is good for large and complex projects.
- Strong approval and documentation control.
- It is suitable for high risk projects, where business needs may be unstable. A highly customized product can be developed using this.

Disadvantages of Spiral Model:

- It is not suitable for small projects as it is expensive.
- It is much more complex than other SDLC models. Process is complex.
- Too much dependable on Risk Analysis and requires highly specific expertise.
- Difficulty in time management. As the number of phases is unknown at the start of the project, so time estimation is very difficult.

Model/Feature	Waterfall	Spiral	Incremental/ Iterative
Specification of All the Requirements in the beginning	Yes	Not all and Frequently Changed	Not all and Frequently Changed
Long term project	Inappropriate	Appropriate	Appropriate
Complex Project	Inappropriate	Appropriate	Appropriate
Frequently Changed Requirements	Inappropriate	Appropriate	Appropriate
Cost	Not costly	Costly	Costly
Cost estimation	Easy to estimate	Difficult	Difficult
flexibility	Not	Flexible	Flexible
Simplicity	Simple	Intermediate	Intermediate
Supporting high risk projects	Inappropriate	Appropriate	Appropriate
Guarantee of Success	High(If requirements are fixed)	High(if changeable)	High(if changeable)
Customer Involvement	Low	Low, After Each Iteration	High, After Each Iteration
Testing	Late	At the end of each phase	After every Iteration
Maintenance	No 8uu	Maintainable	Maintainable
Ease of Implementation	Easy	Complex	Easy

Specifications:
Waterfall:

•Best for static, well-defined projects (e.g., constructing a bridge, software with stable requirements).
•Better if there is a clear vision of the final product and changes are unlikely.

Incremental:

•Suitable for medium-sized projects where some functionality is well-understood and delivering increments adds value.

•Better for projects that can benefit from progressive delivery and feedback.

ERP systems are typically Incremental when focusing on deploying standard modules and Iterative when developing custom solutions or refining systems for unique business processes.
Why Iterative May Be Used:

Iterative Implementation for University ERP Systems

If the university has unique requirements or the ERP system is heavily customized.
When there's uncertainty about the full scope of features required (e.g., specific grading or attendance tracking needs).
To minimize risk by delivering a working product early and refining it.

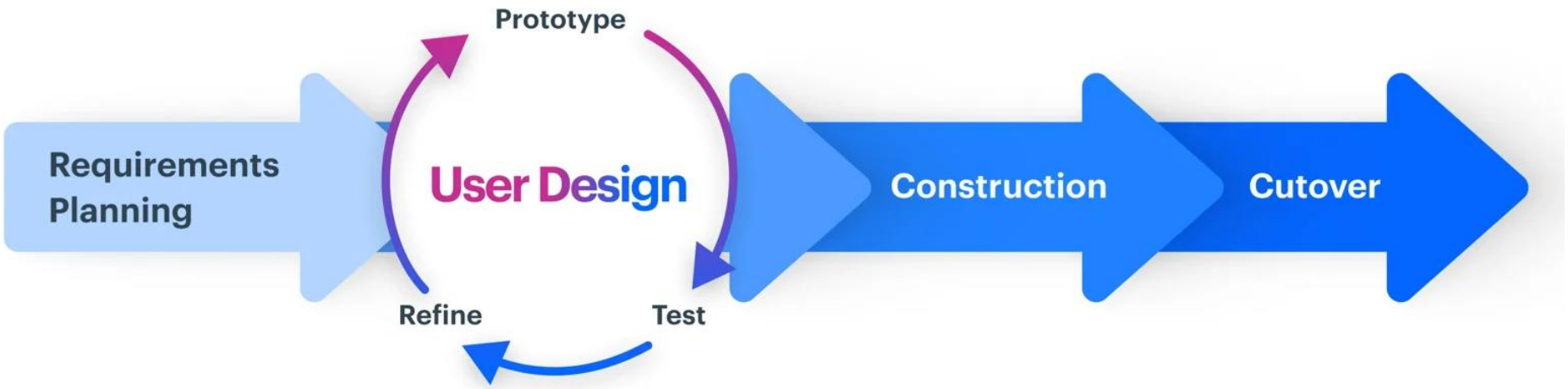
RAD model

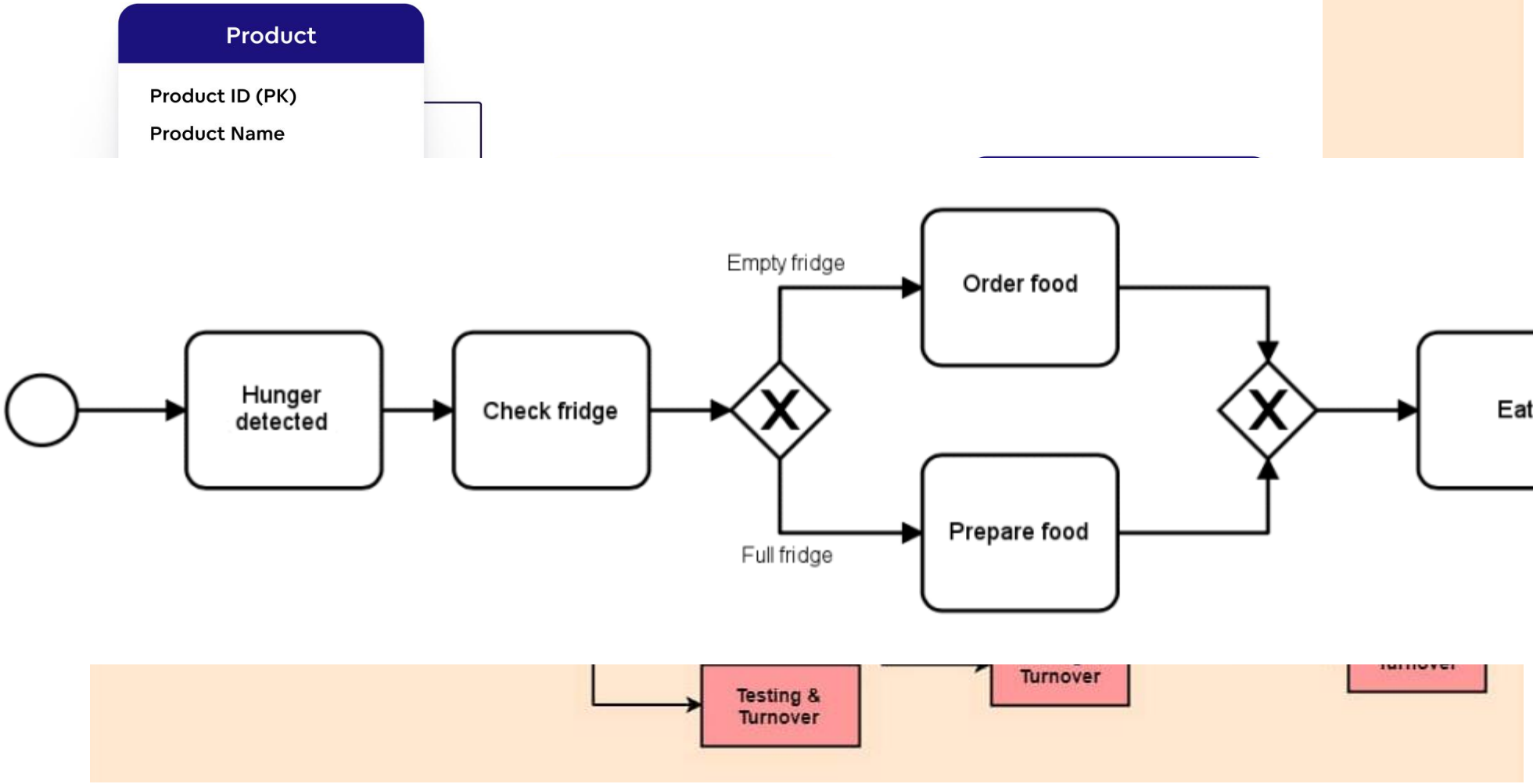
- The Rapid Application Development Model was first proposed by IBM in the 1980s. The critical feature of this model is the use of powerful development tools and techniques.

- RAD is an incremental prototyping approach to software development that end users can produce better feedback when examining a live system, as opposed to working strictly with documentation.

- It puts less emphasis on planning and more emphasis on an adaptive process.

Rapid Application Development (RAD)




Product
Product ID (PK)
Product Name
Empty fridge
Order food
Hunger detected
Check fridge
Eat
Hunger satisfied
Prepare food
Full fridge
Testing & Turnover
Turnover

Advantages:

- The use of reusable components helps to reduce the cycle time of the project.
- Feedback from the customer is available at the initial stages.
- Reduced costs as fewer developers are required.
- The use of powerful development tools results in better quality products in comparatively shorter time spans.
- The progress and development of the project can be measured through the various stages.
- It is easier to accommodate changing requirements due to the short iteration time spans.

Disadvantages:

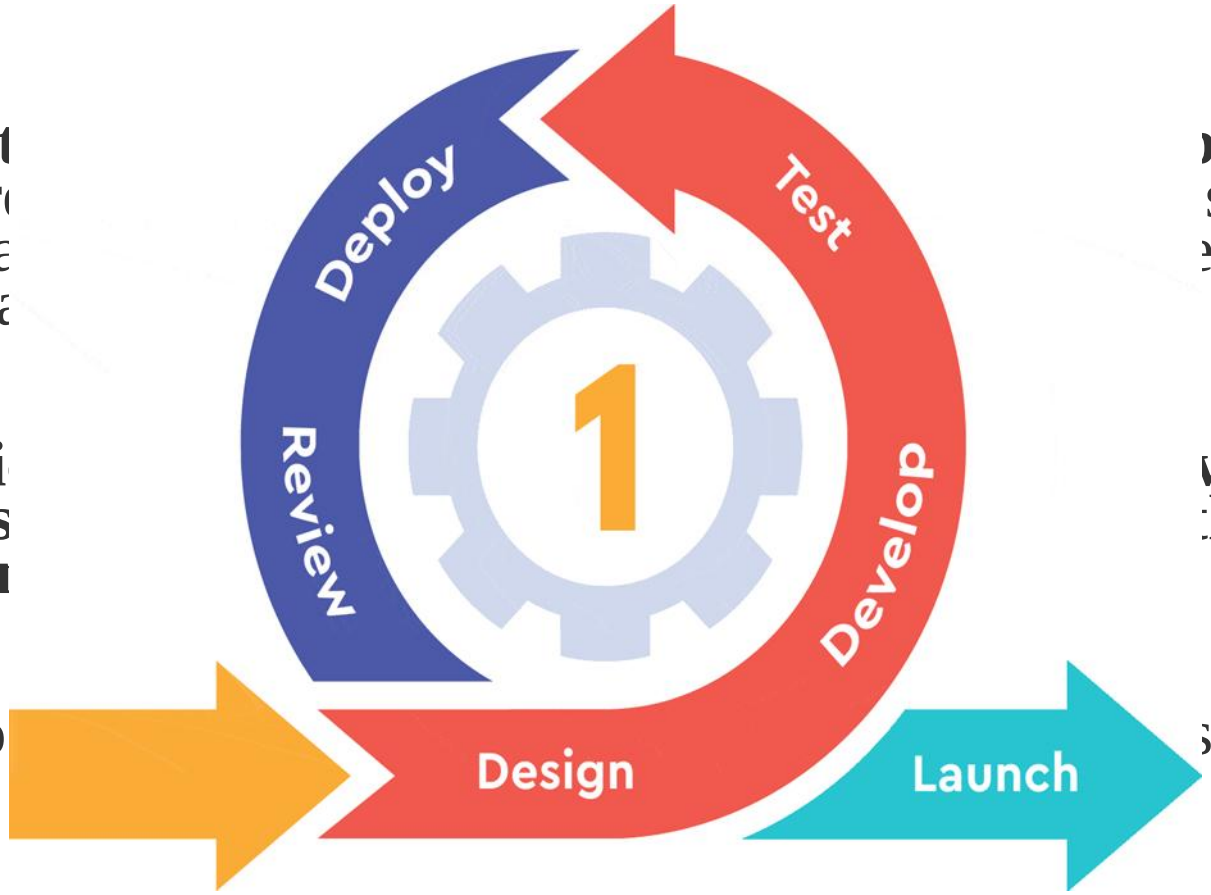
- The use of powerful and efficient tools requires highly skilled professionals.
- The absence of reusable components can lead to the failure of the project.
- The team leader must work closely with the developers and customers to close the project in time.
- The systems which cannot be modularized suitably cannot use this model.
- Customer involvement is required throughout the life cycle.
- It is not meant for small-scale projects as in such cases, the cost of using automated tools and techniques may exceed the entire budget of the project.

RAD MODEL	SPIRAL MODEL
RAD model is a software development model where by the components or functions are developed in parallel as if they were mini projects.	Spiral model is a software development model and is made with features of incremental, waterfall or evolutionary prototyping models.
Rapid development is its main objective.	High assurance is its main objective.
RAD model requirements and early stage planning is not necessary.	Spiral model requirements and early stage planning is required.
It is necessary to have detailed documentation but in a limited manner.	Detailed documentation is required.
Requirements are specified as time boxed release manner.	Requirements are specified in the beginning.

RAD model is used between large and medium project.	Spiral model is used for large project.
There is low amount risk in RAD model.	There is medium to high amount risk in spiral model.
In RAD model small team size is required.	In spiral model large team is required.
Flexibility to change in RAD model is Easy.	Flexibility to change in spiral model is not that difficult.
In RAD model overlapping of phases is possible.	In spiral model overlapping of phases is not possible.
Testing is done in RAD model after completion of coding.	Testing is done in spiral model at the end of the engineering phase.
Cost of RAD model is Low.	Cost of spiral model is very expensive.
Customer involvement is only at the beginning.	Customer involvement is high as compared to RAD model.
RAD model is having easy maintenance.	Spiral model has typical maintenance.
It offers reusability.	Reusability is possible to some extent.

Agile Model

- **The agile met** **pproach**. Rather than treating requir steps, an agile model makes them a nt from developers, management, a
- **Work is typi wn as "sprints," in which the res heir customers and perform testi**
- Agile tends to startups, where speed and flexibility



Advantages of Agile Methodology :

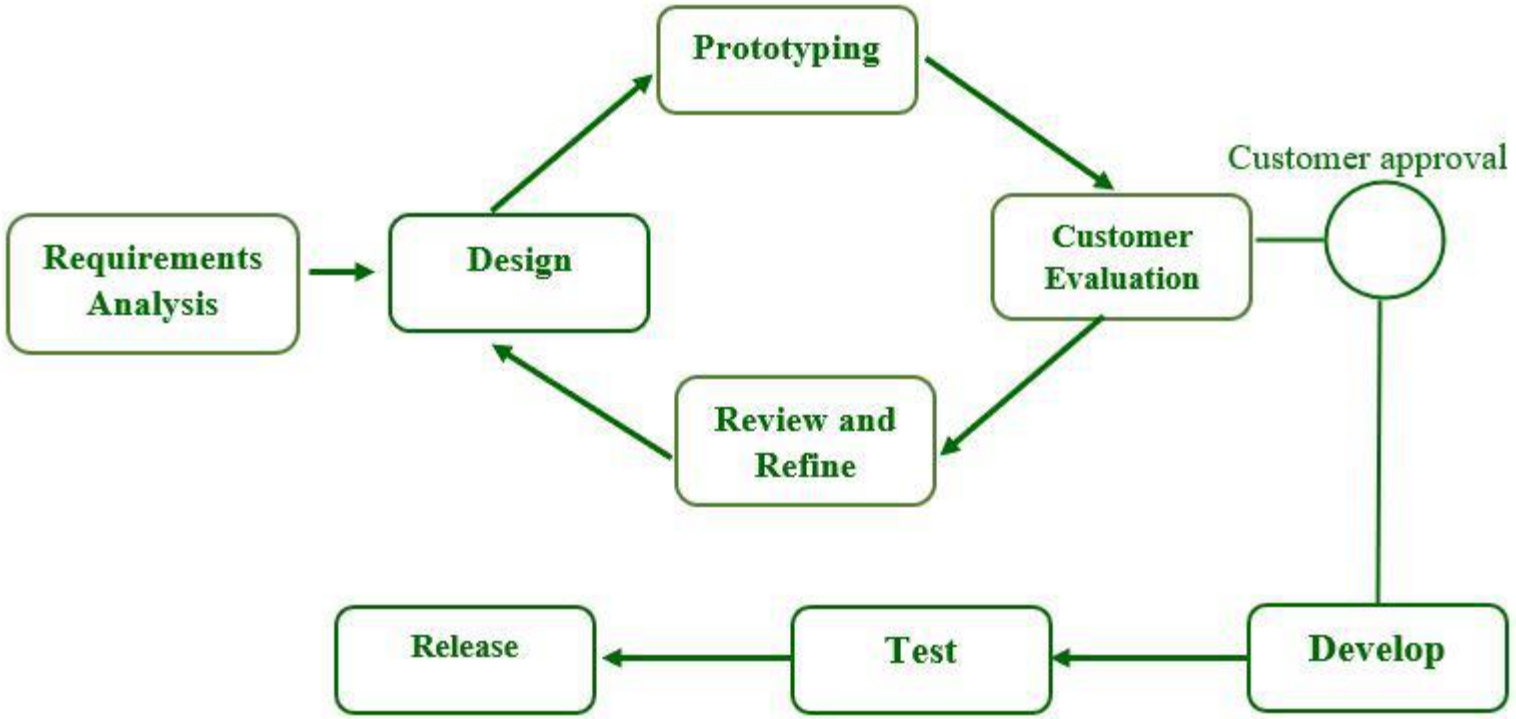
- The customers are satisfied because after every **sprint** working feature of the software is delivered to them.
- In Agile methodology the daily interactions are required between the business people and the developers.
- In this methodology attention is paid to the good design of the product.
- Changes in the requirements are accepted even in the later stages of the development.

Disadvantages of the Agile Methodology :

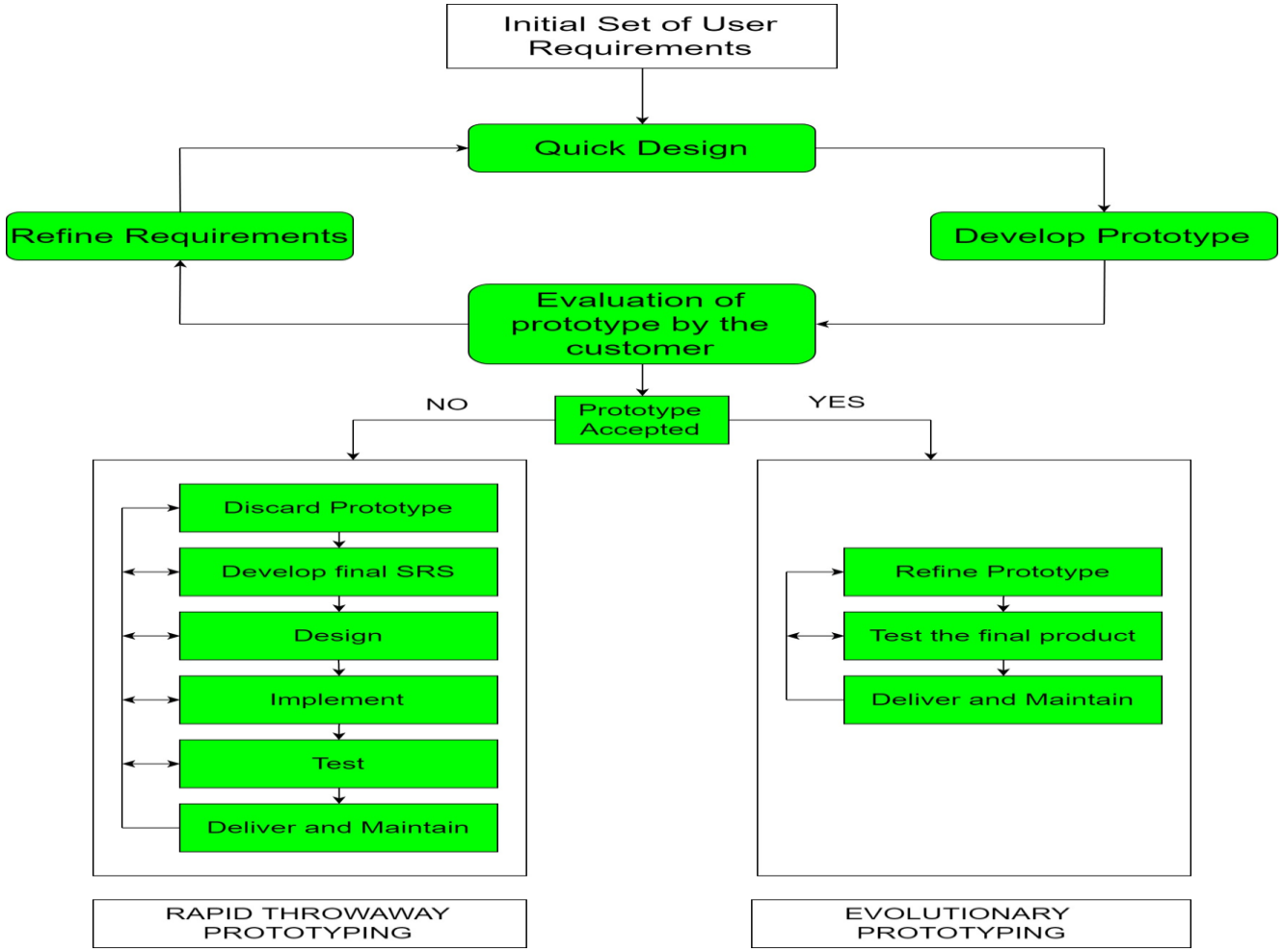
- In Agile methodology the documentation is less.
- Sometimes in Agile methodology the requirement is not very clear hence it's difficult to predict the expected result.
- For complex projects, the resource requirement and effort are difficult to estimate.

What is Prototyping

- **Prototyping** is an experimental process where design teams implement ideas into tangible forms from paper to digital.
- This model is used when the customers do not know the exact project requirements beforehand.
- Prototyping is defined as the process of developing a working replication of a product or system that has to be engineered.
- In this model, a prototype of the end product is first developed, tested, and refined as per customer feedback repeatedly till a final acceptable prototype is achieved which forms the basis for developing the final product.


Prototyping
Customer approval
Requirements Analysis
Design
Customer Evaluation
Review and Refine
Release
Test
Develop

The Prototyping Model



Advantages :

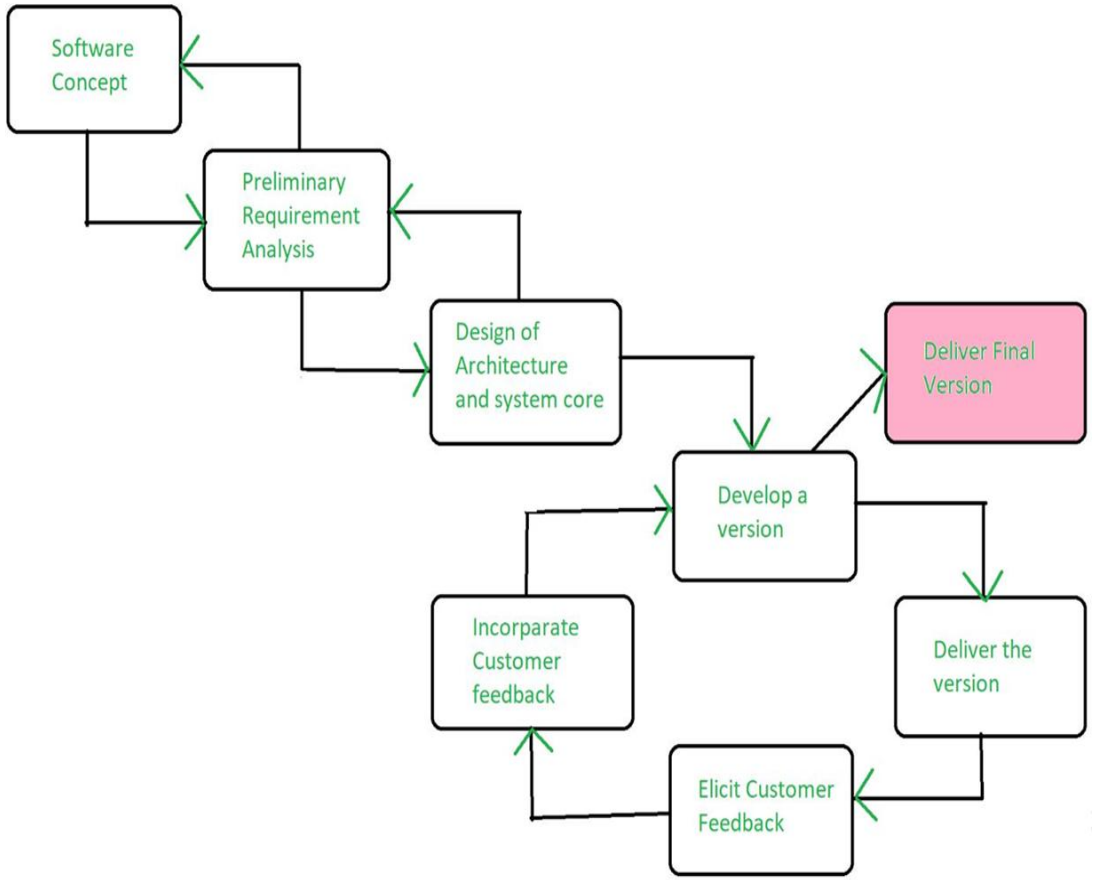
- It can be used by the developer for more complicated projects in the future.
- It ensures a greater level of customer satisfaction and comfort.
- It helps developers and users both understand the system better.

Disadvantages:

- This model is costly.
- It has poor documentation because of continuously changing customer requirements.
- There may be too much variation in requirements.
- Customers sometimes demand the actual product to be delivered soon after seeing an early prototype.

Evolutionary Model

- The evolutionary model of the software development life cycle combines the incremental and iterative models.
- The actions carried out in this paradigm include deploying the system in a big bang release or incrementally over time.



- It is better for software products that have their feature sets redefined during development because of user feedback and other factors.

- The development cycle is divided into smaller, incremental waterfall models by the evolutionary development model, where users can access the product at the end of each cycle.

- According to the evolutionary approach, work should be divided into smaller portions, given a priority list, and then sent to the customer one at a time.

- Both the number of delivery to the consumer and the number of portions are enormous.

Advantages:

- The evolutionary model's main benefit is that it helps build customer confidence and allows the user to verify each step of the development process.
- Due to every module is clearly visible, there are fewer opportunities for errors.
- It aids in reducing software project risk.
- It reduces development costs.
- It reduces the number of major issues during testing.

Disadvantages:

A variety of modifications made by clients throughout development may cause a delay in the final program delivery.

It is challenging to break the problem up into manageable chunks that the client would accept and that could be gradually executed and provided.

Software Engineering | Software Process Framework

- Tasks: concentrate on a small, particular goal.
- An action is a group of operations that result in a significant output.
- Activities are a collection of linked tasks and actions toward a main goal.

Umbrella activities:

- Umbrella Activities are that take place during a software development process for improved project management and tracking.

Umbrella activities:

- **Software project tracking and control:** This is an activity in which the team can assess progress and take corrective action to maintain the schedule. Take action to keep the project on time by comparing the project's progress against the plan.

- **Risk management:** The risks that may affect project outcomes or quality can be analyzed. Analyze potential risks that may have an impact on the software product's quality and outcome.

- **Software quality assurance:** These are activities required to maintain software quality. Perform actions to ensure the product's quality.

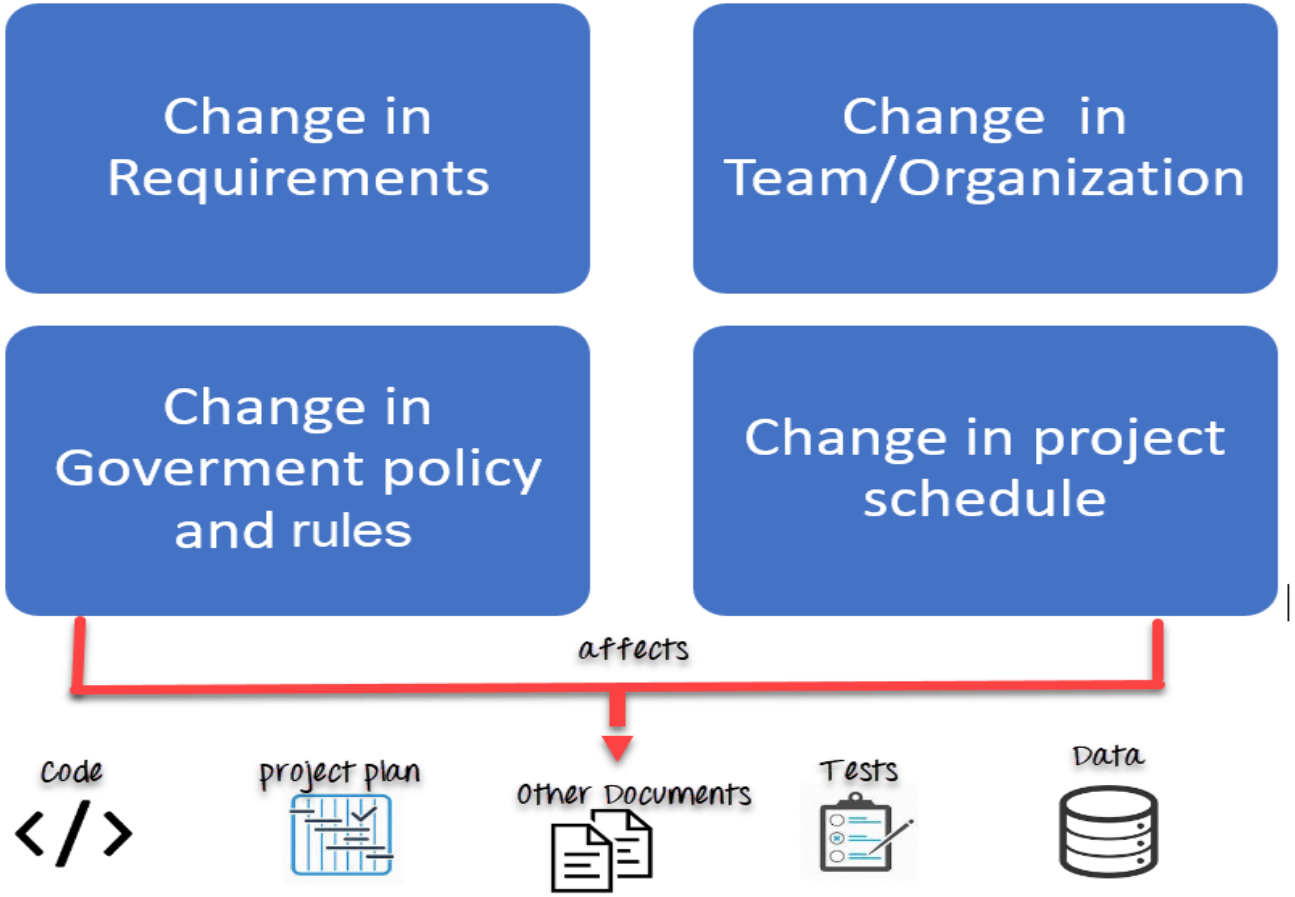
- **Formal technical reviews:** It is required to assess engineering work products to uncover and remove errors before they propagate to the next activity. At each level of the process, errors are evaluated and fixed.

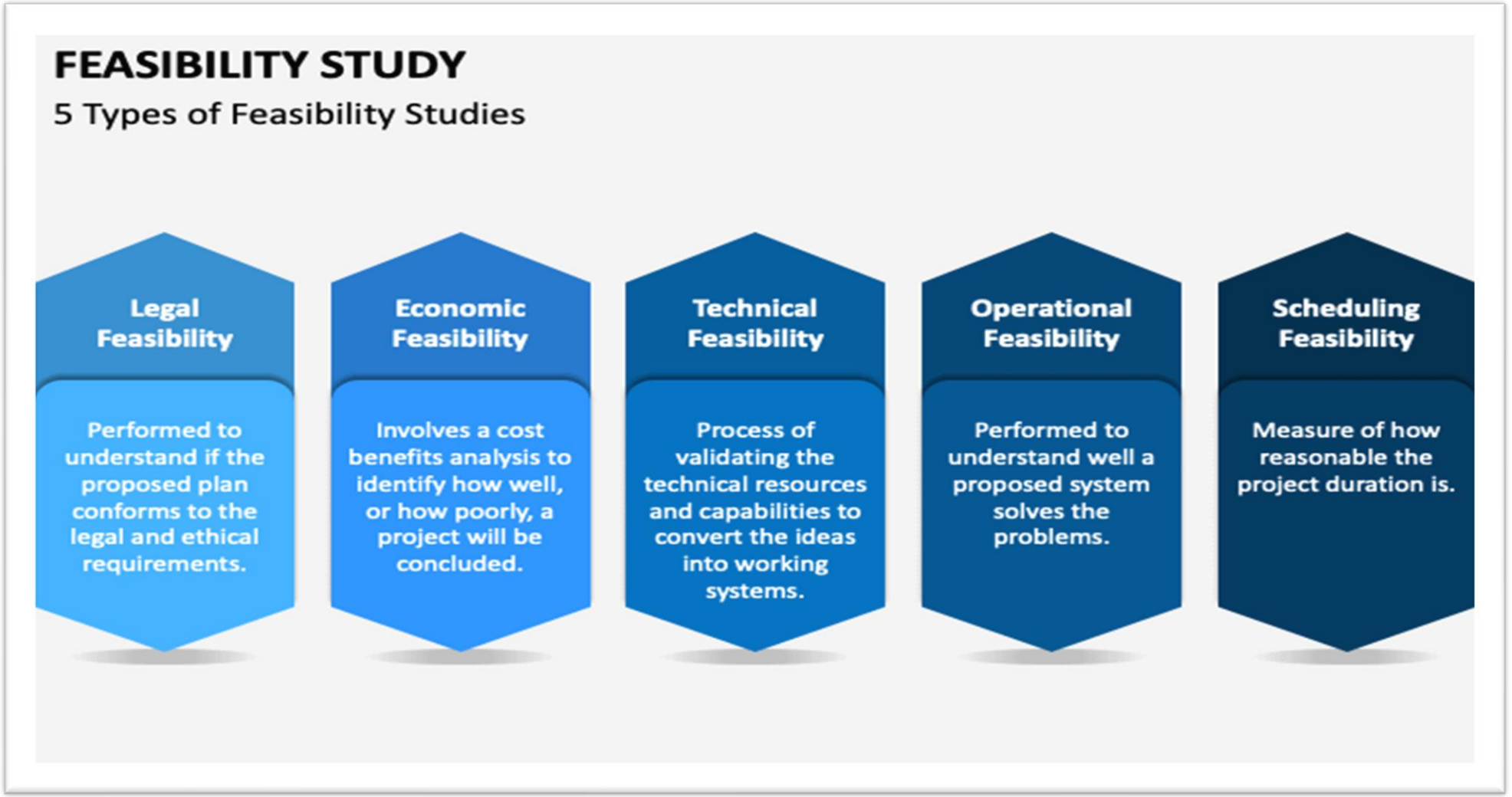
- **Software configuration management:** Managing of configuration process when any change in the software occurs.

- **Work product preparation and production:** The activities to create models, documents, logs, forms, and lists are carried out.

- **Reusability management:** It defines criteria for work product reuse. Reusable work items should be backed up, and reusable software components should be achieved.

- **Measurement:** In this activity, the process can be defined and collected. Also, project and product measures are used to assist the software team in delivering the required software.


Change in Requirements
Change in Team/Organization
Change in Goverment policy and rules
Change in project schedule
affects
Code
project plan
Other Documents
Tests
Data


FEASIBILITY STUDY
5 Types of Feasibility Studies
Legal Feasibility
Performed to understand if the proposed plan conforms to the legal and ethical requirements.
Economic Feasibility
Involves a cost benefits analysis to identify how well, or how poorly, a project will be concluded.
Technical Feasibility
Process of validating the technical resources and capabilities to convert the ideas into working systems.
Operational Feasibility
Performed to understand well a proposed system solves the problems.
Scheduling Feasibility
Measure of how reasonable the project duration is.


Thank you